

Synthetic Daily CPI Tracker Methodology

This document outlines the end-to-end methodology used to construct the daily synthetic Consumer Price Index (CPI) tracker for Bolivia. The pipeline operates autonomously, pulling raw supermarket data (scraped by Mauricio Foronda), applying AI-driven categorizations, and calculating a mathematically robust, chained Jevons index that mirrors official INE (Instituto Nacional de Estadística) structures. The method was adapted from Alberto Cavallo’s 2013 paper: “Online and official price indexes: Measuring Argentina’s inflation.”

1 Data Ingestion & Preprocessing

The tracker relies on daily web-scraped price data from Bolivia’s largest supermarket chain (30+ locations) with stores in **La Paz, Santa Cruz, and Cochabamba**. The retailer has over 28,000 unique products in many different categories.

- **Frequency:** Daily data extraction.
- **Sanitization:** Raw identifiers (`id_producto`) are stripped of special characters, lowercased, and standardized using a `normalize_id` function to prevent duplication.
- **Temporada Filtering:** Any product explicitly tagged as seasonal (`temporada`) in its metadata or name is structurally excluded. This mitigates excessive noise and extreme price volatility that is not representative of core inflation trends.

2 Product Categorization & AI Mapping

To align supermarket SKUs with official macroeconomic data, each product is mapped to the standard **Classification of Individual Consumption According to Purpose (CCIF)**. An initial mapping of the 28,000 products was done by an AI-coded application trained on the Instituto Nacional de Estadística’s CCIF.

- **Master Mapping:** A persistent historical mapping dictionary (`mappings/Final_Complete_Categories.csv`) is loaded into memory to categorize known products instantaneously.
- **AI-Driven Discovery:** When the ingestion pipeline encounters an unseen product ID, it extracts the product name and queries the **Gemini 2.5 Flash** model.
- **Context Injection:** The model is provided with the official *INE 2016 Methodology PDF* and the *CCIF categorization guidelines* as strict context. It is forced to output a JSON payload classifying the new item into one of the designated **core categories** (or “Unmapped” if inapplicable).

- **Persistent Storage:** The AI’s decisions are appended back to the master CSV, ensuring the API is only queried for *novel* products, drastically reducing operational costs.

3 Core Basket & Weights

The synthetic CPI **does not** cover all 12 INE categories because the supermarket does not sell products related to all categories, such as housing or transportation. Instead I define a “Core-5” basket, which gives 49.13% coverage of the official CPI.

Table 1: Core-5 Category Weights

Category	Raw Weight	Normalized Weight
Alimentos y Bebidas No Alcohólicas	27.06	55.08%
Prendas de Vestir y Calzados	7.56	15.39%
Bienes y Servicios Diversos	7.55	15.37%
Muebles, Bienes y Servicios Domésticos	6.08	12.38%
Bebidas Alcohólicas y Tabaco	0.88	1.79%

(Note: Raw weights are normalized out of their sum (49.13) so the resulting sub-basket sums to 100%).

4 Index Calculation

The core calculation relies on a **Chained Jevons Index**, the standard geometric formula used by Cavallo (2013) for elementary aggregate inflation where product substitution is likely.

Step A: Day-Over-Day (DOD) Relatives

For every active product, the daily relative price change is calculated against its immediate previous observed price:

$$RelativeDOD_{i,t} = \frac{Price_{i,t}}{Price_{i,t-1}}$$

(Missing prices are omitted from the daily relative calculation, ensuring the index only moves when a price change is actively observed).

Step B: Attrition and Imputation Limit (40-Day Rule):

The pipeline enforces a strict 40-day inclusion limit. If a product is unobserved for more than 40 consecutive days, it is considered permanently dropped from the matched-model sample. If the exact same product reappears on day 41 or later, its day-of-return relative is nullified. The product simply establishes a new base price for the following day, preventing artificial index spikes from severely obsolete baseline prices while keeping the overall CPI structurally intact.

Step C: Elementary Jevons Aggregate

Within each city (c) and category (k), the elementary index for day t is the geometric

mean of all constituent product relatives:

$$JevonsMean_{c,k,t} = \exp \left(\frac{1}{n} \sum_{i=1}^n \ln(RelativeDOD_{i,t}) \right)$$

Step D: Time-Series Chaining

The daily geometric means are chained sequentially using a cumulative product to construct a continuous time series anchored at 100:

$$ChainedIndex_{c,k,t} = ChainedIndex_{c,k,t-1} \times JevonsMean_{c,k,t}$$

5 Aggregation

Once the elementary category indices are calculated, they are aggregated upwards to form macroeconomic indicators.

City-Level Aggregation

For a given city, the overall CPI is the weighted arithmetic average of its 5 core category indices using the normalized weights established in Section 3:

$$CityCPI_{c,t} = \sum_k (ChainedIndex_{c,k,t} \times NormalizedWeight_k)$$

National Aggregation

The National CPI is computed as the weighted arithmetic average of the three metropolitan indices, based on official demographic/economic distributions:

- **Santa Cruz:** 41.47%
- **La Paz:** 39.10%
- **Cochabamba:** 19.43%

$$NationalCPI_t = \sum_c (CityCPI_{c,t} \times RegionalWeight_c)$$

6 Daily Forward-Filling & N-Counts

- **Imputation:** To guarantee a continuous daily time series (even on holidays where scrapers may fail), final index values are forward-filled (`ffill`).
- **N-Counts:** The pipeline simultaneously counts the number of valid observations per category per day. This metadata is saved to `supermarket_1_daily_n_counts.csv` to monitor sample size health and identify data degradation.

7 Automation Infrastructure

The entire pipeline runs via a **GitHub Actions Cron Job** at 12:00 UTC (8:00 AM local time) daily. The runner instantiates a Python 3.10 environment, triggers the AI mapping for novel SKUs, runs the vectorized calculations, and seamlessly commits the updated historical `.csv` and `.json` artifacts back to the repository.

8 Hipermaxi Subcategories by Record Volume

Table 2: Hipermaxi Subcategories by Record Volume

Subcategory	Record Count
Skin Care	1,800,677
Hair Care	1,336,777
Basics / Staples	884,422
Coffee & Substitutes	695,349
Wines & Champagne	649,930
Condiments & Legumes	636,724
Cookies & Crackers	607,444
Chocolates & Candies	593,662
Laundry Care	566,359
Snacks	559,747
Liquors & Spirits	551,504
Infant Nutrition	535,377
Home Accessories	493,899
Oral Care	456,611
Various Candies	445,672
American Products	433,773
Cold Cuts (External Brands)	404,715
Diet & Healthy	402,618
Dressings	398,756
Baking / Pastry	370,956
Bathroom Cleaners	360,144
Living Room & Closet Cleaners	348,292
Baby Care	305,419
Frozen Meats	305,036
Ice Cream, Cassatas & Popsicles	299,267
Family Care	290,566
Kitchen Cleaners	264,906
Canned Meals	264,748
Jams & Honeys	248,437
Cheeses	241,689
Canned Vegetables	239,229
Pets	230,052
Juices & Nectars	225,002

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Subcategory	Record Count
Yogurts	212,092
Glassware / Crystal Housewares	204,709
Milks	199,325
Sodas / Soft Drinks	195,050
Bakery (External Brands)	194,973
Kitchenware	186,621
Diapers	170,455
Doughs & Frozen Meals	169,142
Various Beverages	155,777
Cold Cuts (Hipermaxi Brand)	147,403
Kids	143,045
Stationery	141,491
Cigarettes	125,317
Pastry (External Brands)	124,384
Canned Fruits	120,466
Plastic Housewares	116,576
Party Supplies	114,703
Waters	111,649
Costume Jewelry / Personal Care	110,458
Pets General	108,621
Sports / Isotonic Drinks	107,422
Beers	107,007
Hardware	105,344
Vegetables	104,506
Powdered Drinks	102,804
Sauces & Tomato Paste	97,128
Living Room & Bedroom	93,113
General Kitchen	87,485
Seasonal: Christmas	81,211
US Beverages	80,837
Imported Products	75,241
Butters & Margarines	72,657
Frozen Pulps & Fruits	70,976
Eggs	67,046
Beef	64,295
BBQ / Grill Meats	64,222
Insecticides	61,994
Adult Care	61,457
Traditional Doughs (External)	58,808
Bakery (Hipermaxi Brand)	55,879
Fruits	52,593
Shoe Care	50,406
Energy Drinks	47,729
Automotive	47,003

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Subcategory	Record Count
Gifts & Decorations	43,944
Toys: Boy / Baby	43,709
Portioned / Repackaged (Hipermaxi)	36,206
Pastry (Hipermaxi Brand)	34,164
Seasonal	30,547
Chicken	30,019
Desserts	27,175
Porcelain / Ceramic Housewares	25,166
Gourmet	24,406
Dulce de Leche & Manjar	23,355
Toys: Girl / Baby	22,469
Bakery Specialties (Hipermaxi)	20,235
Educational Toys / Seasonal	20,153
Vegetables (Delifresh)	17,276
Processed Meats	16,606
General Bathroom	13,829
General Camping	13,374
Milk Cream / Heavy Cream	12,935
Home Appliances / White Goods	12,866
Vacuum-Packed Meats (External)	12,705
Seasonal: Halloween	11,978
Special Offers	11,574